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Electric double-layer of [emim][DCA] ionic liquid at heterogeneous interface of TiO_2/C composite: From simulation to experiment

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The hydrophilicity/hydrophobicity of heterogeneous interface of the electrode material is critical for the performance of ionic liquids supercapacitors. However, the behavior of electrolytes at the electrode interface, especially the interface microstructure and properties is not yet clear. We investigated the electric double-layer of [emim][DCA] on slit pore of TiO_2/C composite with the heterogeneous surface by combining molecular dynamics simulation and experimental measurements. The results of simulation show that coverage ratio of carbon on the surface affect the number density distribution, orientation and diffusion coefficient of [emim][DCA]. Moreover, we calculated the charge distribution of electric double-layer of [emim][DCA] and the capacitance. The results indicate that the capacitance of slit pore with the surface of TiO_2 with 78.7% carbon coverage ratio has the highest capacitance at high surface charge density. We performed an electrochemical experiment using composite of TiO_2 covered with carbon as the electrode material, and obtained the capacitance from cyclic voltammetry (CV). The experimental results also indicate there is an optimal coverage ratio of carbon, which achieves the highest capacitance. Our researches could provide a microscopic perspective of the electric double layer at the heterogeneous interface and the guidance for the design of electrode material.

1. Introduction

Ionic liquids (ILs) are used as electrolytes in electric double-layer capacitors (EDLC) since they have low vapor pressure, high electrochemical stability and energy density. For a given IL, the surface properties of solid electrode materials have a great influence on the interaction between the ILs and the surfaces. The properties of electric double-layer depend critically on the interface of ILs and the surface of electrode, which has driven increased interest in the behavior of ILs at the solid interface[1] and in confined geometries structure of electrode material recently [2]. When the interaction between the ILs and substrate is strong, interference from contacting the surface has multifold influence on the static arrangement and dynamic behavior of ILs[3], including over screening [4] and templating effects [5].

Researchers have studied the effect of geometric heterogeneity of electrode surface on properties of electric double-layer. Lu et al. [6]. found the ion layers can be extended deeper into the bulk electrolyte solution by the stronger interaction of the rough electrode. Due to the greater effective contact area between the ions and electrode, the rough electrode possesses a larger capacitance than the non-rough one. Hu et al. [7]. discussed the influence of the electrode surface and electrolyte structure on electrode capacitance and EDL structure. However, the role of the hydrophilic/hydrophobicity heterogeneity structure of electrode surface on EDL is unclear and perhaps controversial. Wang et al.[8]. reported the unique 3D pore structure of N-doped microporous carbon and the N-doped carbon electrodes exhibit high specific capacitance and good cycling stability in ionic liquid electrolyte. Ewert et al. [9]. reported a higher N-doping of the electrode material improves the symmetric full cell capacitance of the match and considerably increases its long-term stability. The results of Sathish et al. [10]. also show the amount of N-doping varies with the nature of the N containing organic compounds and the supercapacitance behaviour depends on the amount of N-doping as well as the nature of N-doping in the graphene. Yan et al. [3]. observed that hydroxyl group on the surface and Br formed H-bond which results in the liquid like ordering and breaks down the electrostatic network in ILs.

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Chen et al. [11]. synthesized a NiO/CNTs composite and can be used as electrode which exhibited good capacitive behavior and cycle ability. The above research results show that the chemical heterogeneous on the surface of the electrode will affect the electric double layer structure and improve the capacitance performance of the electrode.

It is important to obtain a thorough understanding of the electrical double layers (EDLs) at the interface of ILs and electrodes, because the microstructure and capacitance of these EDLs play an essential role in determining the system performance. However, the complex heterogeneity and intermolecular interactions at the electrode interface challenge the description of electrochemical behaviors by common experiment. Molecular simulation has great advantage on understanding the complex structure of ionic layers in the EDLs. The EDLs structure of the ILs has been discovered in various molecular simulations. Jo et al. [12]. found that differential capacitance behavior mainly stems from the ion exchange between EDL and bulk region by molecular dynamic (MD) simulation. Liu et al. [13]. reported that ILs with FSI^- and Tf^- anions show substantially different EDL structures and differential capacitance (C_d) features by MD simulation. Wang et al. [14] focused on the effect of force field to model the Au(111)/ionic liquid interface. Bedrov et al. [15]. demonstrated how atomistic molecular dynamics simulations provide a powerful tool for dealing with those challenges in understanding of EDL properties and their influence on the performance of supercapacitors. And it can facilitate the design of novel materials for advancing energy storage technologies. Therefore, the MD simulation should have the ability to reflect the intrinsic characteristics of the surface structure. Moreover, Ma et al. [16]. bridged the gap between macroscopic electrochemical measurements and microscopic molecular dynamic simulations for porous carbon supercapacitor with ILs. And they explored the path of implementing the techniques of cyclic voltammetry and electrochemical impedance spectroscopy using MD simulations. From the above research results, we can see that molecular simulation is a good tool for analysis in nanoscale, which can reflect the influence of surface properties from a molecular lever. Researchers have corresponded simulation results with electrochemical measurements.

[emim][DCA] is a low viscosity imidazolium-based ionic liquid. The low viscosity provides higher mobility of charges that is significantly observed in ionic conductivity measurement. Obviously, it is a beneficial condition for conductivity enhancement[17]. Supercapacitor cells with the [emim][DCA] electrolyte exhibited a high specific capacitance (76.4 F g^{-1}) and high specific energy (19.8 Wh.kg^{-1}) [18]. Carbonaceous materials are widely used as electrode materials for supercapacitors because they have good conductivity and large specific surface area[19,20]. EDLCs consisting of graphene have shown superior performances compared to conventional capacitors. For example, Shaikh et al. [21] synthesized a three-dimensional nanoporous honeycomb sulfur-embedded graphene with a large 3.2 V at a constant applied current density of 3.2 A g^{-1} Potential window and high energy density of 124 Wh.kg^{-1}. Docampo-Alvarez et al. [5] studied the mixtures of alkylammonium-based protic ILs and alkylmethylimidazolium-based aprotic ILs by means of molecular dynamics simulations. Jo et al. [12] researched the effects of cyano-group in ILs on differential capacitance in graphene supercapacitors. Fang et al. [22] demonstrated that relative ion-electrode van der Waals interactions play an important role in the population of ions adsorbed in the first interfacial layer near uncharged electrodes. Antenor et al. [23] investigated seven GO compositions with different levels of oxygenation by MD simulations. They observed that the high oxidation degree obstructs the formation of new hydrogen bonds, which considerably affects their hydration properties. The results provide a physical background for applications of GO in electrodes of supercapacitors. However, the hydrophobicity of carbon materials is not conducive to the adsorption of ILs on the surface, and unfavorable to the diffusion of ILs in the pore of porous materials. TiO_2 is a chemically stable material and is readily and inexpensively commercially available used as electrodes in electrochemical devices [24]. Highly ordered TiO_2 nanotube arrays hold great promise as supercapacitor electrodes[25]. However, the relatively small specific capacitance of TiO_2 was attributed to the poor electrical conductivity. TiO_2 is hydrophilic, but its conductivity is lower than that of carbon. Therefore, we adopted the method of partially covering carbon on titanium oxide materials to improve the hydrophilic and hydrophobic properties of materials without damaging the conductivity of electrode materials. In our previous work[26], the TiO_2/C composites show their typical capacitance behavior and fast ion response with aqueous electrolyte. A higher specific volumetric capacitance was obtained in the composite TiO_2/C than the value of Vulcan XC-72. And the long time performance of the TiO_2/C composite is better than that of Vulcan XC-72, suggesting a more stable interface between TiO_2/C and electrolyte in the porous structure of the composite. Based on this, we have reason to believe that TiO_2/C composite materials will have high performance of capacitance with ionic liquid electrolyte.

In this paper, molecular dynamic simulation was used to investigate the microstructures of [emim][DCA] in the pore with heterogeneous surface of TiO_2/C and the characteristics of EDL. The effect of surface heterogeneity and different ratio of C coverage on capacitance were discussed through investigation of the electric double-layer of [emim][DCA] on slit pore with heterogeneous surface of TiO_2/C. At the same time, TiO_2/C composite was synthesized as electrode material to mimic the surface heterogeneous electrode material used in simulation, and its capacitance was investigated by performing electrochemical experiment, and obtained the capacitance from cyclic voltammetry (CV) to verify the predicted results of molecular simulation.

2. Simulation and experimental methods

2.1. Models and simulation details

2.1.1. The structure model of simulation system

There are two different electrode model settings in the simulation works. A more natural simulation would be a methodology that allows simulation of electrolyte confined between two electrodes[15]. However, the simulation based on this model requires a lot of computing time because the distance between positive and negative electrodes should be large enough. An alternative model is to simulate positive and negative electrodes separately. Chen et al. [27] compared the results of single-electrode model and two-electrode model studies, indicating that the single-electrode model calculation is more time-saving. And the results are similar to those of two-electrode model. Another problem is related to how the charged surface is coupled with the rest of the system in MD simulation. The most straightforward approach is to confine electrolyte between two solid surfaces and then apply constant charges + Q and −Q on the electrodes[15]. Li et al.[28] found that it is advantageous to use a constant potential when the electrode surface is porous or rough. When the electrode is a flat plate and the ionic liquid is at a potential between −3 V and 3 V, the result of using a constant potential or a constant surface charge is the same. Based on previous research results, we built single electrode models to simulate the positive and negative electrodes, respectively. Constant surface charge is applied on the electrodes. We applied 0 C/m^2, 0.06445 C/m^2, -0.06445 C/m^2, 0.3222 C/m^2, -0.3222 C/m^2 charge density to the surface of slits by applying certain charges to the carbon atoms. The structure model of simulation system is shown in Fig. 1a. The simulation box size is $5.922 \times 6.489 \times 3.675$ nm^3, and the width of the slit is 2.0 nm. Fig. 1b is the molecular structure of [emim][DCA]. Composite TiO_2/C is modeled by coated with graphene sheet and Fig. 1c shows the graphene sheet covered on the surface of TiO_2 with different coverage ratio of 49.5%, 78.7%, 100% from left to right, respectively.

2.1.2. The force field model and simulation details

The LAMMPS[29] package is used to perform simulation, and the NVT ensemble simulation was used to determine the number density and diffusion of [emim][DCA] in each case. Firstly, all simulated systems were relaxed at 800 K for 1 ns, and then cooled from 800 K to 398 K for 1 ns, after that simulations run 1 ns at 398 K. In order to relax the energy in a short period of time, we first raised the temperature and then lowered the temperature. This treatment is to speed up the energy relaxation process of ILs without any adverse effect on the final results. And it has been confirmed by many researchers[27,30,31]. NVT simulations were then performed for an additional 20 ns, saving the data every 1 ps, the second half of the trajectories were used for analysis. The periodic boundary condition (PBC) was applied in three dimensions. Nose-Hoover thermostat method[32] was used for temperature control. The Newton's classical equation was used to describe the motion of atoms, and the equation was solved using the velocity-Verlet algorithm[33] with a step size of 1 fs. The particle-particle particle-mesh method with a real space cut-off of 1.0 nm was used to calculate the electrostatic interaction[34]. The positions of all the atoms of graphene and TiO_2 were fixed throughout the simulation process.

The force field model and parameters of ionic liquid [emim][DCA] and the pair coefficients of all carbon atoms in graphene were taken from Atilhan et al.'s work [35], and the parameters of TiO_2 were from Mohammadpour et al.'s work [36]. The simulation of transitions metals is somehow controversial due to the relevance of the electronic degrees of freedom in the behavior of the metal atoms in the condensed phase, and the simulation of transition metal is a difficulty. Considering that the classical molecular dynamics simulation we used in this manuscript only calculates van der Waals force and electrostatic force. We think OPLS-AA force field is enough to reflect the surface heterogeneity. For different atoms, Lorentz-Berthelot mixed rule was used to calculate L-J interaction parameters. The diffusion coefficient is calculated by Einstein equation.

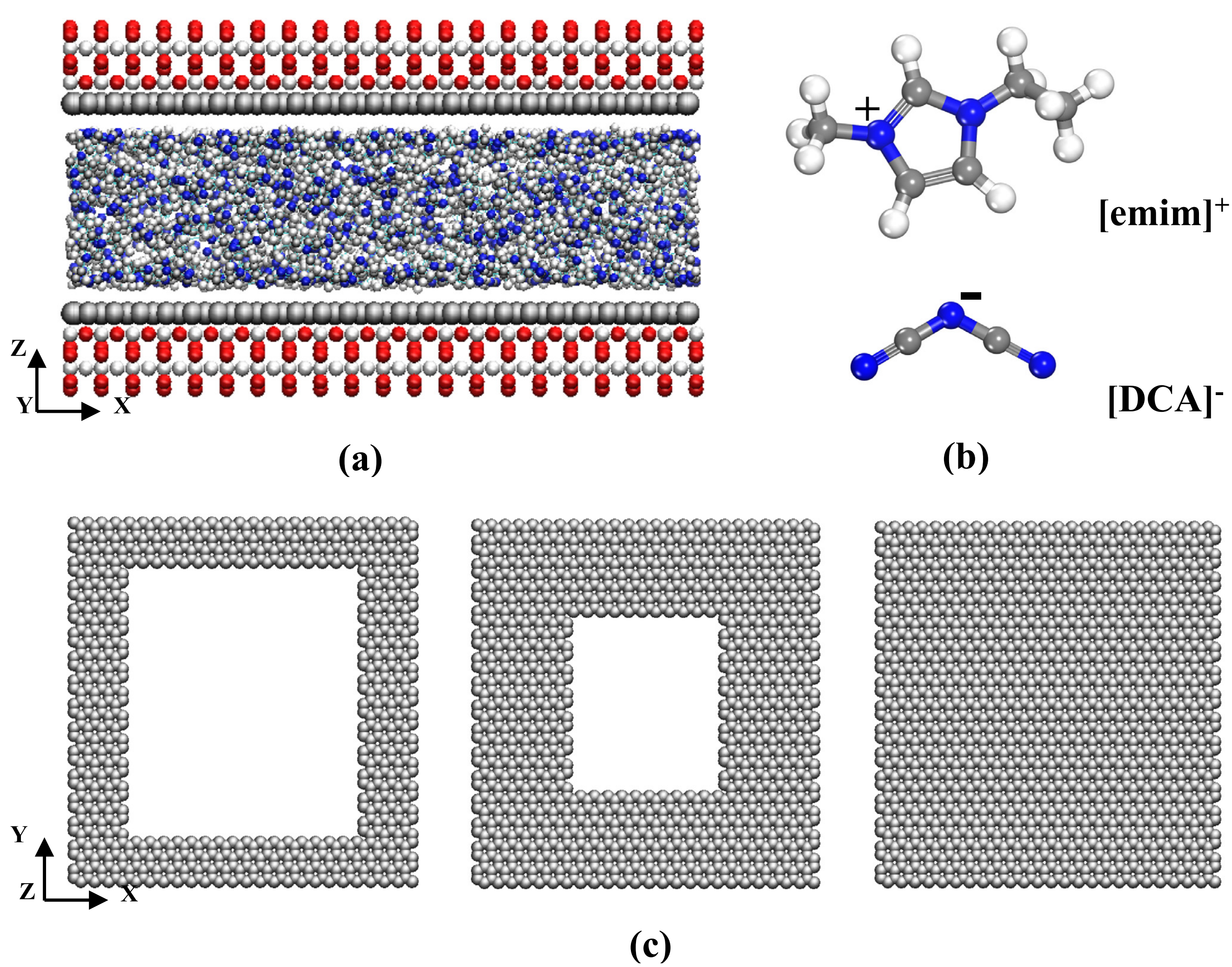

Fig. 1. (a) The structure model of simulation system. (b) The structures of cation of $[emim]^+$ and anion of $[DCA]^-$. (c) Graphene covered on TiO_2 with coverage ratio of 49.5%, 78.7%, and 100% from left to right, respectively. (In the Figure, red, silver, blue, gray, and white balls represent titanium, oxygen, nitrogen, and hydrogen atoms, respectively.)

2.1.3. Initial configuration of simulation box

The number density of ILs in slits is often different from that in bulk phase. The number of cations and anions is often different near the electrode surface. In order to obtain the initial configuration, we constructed a relatively large simulation box as shown in Fig. 2, the number density of [emim][DCA] here is the density of bulk ILs [37]. The slit was placed in the middle of box. A total of 800 pairs of [emim][DCA] were put into the box. After running for 10 ns under NVT ensemble, the number of anions and cations in the slit was counted to determine the number density of ILs and the number of anions and cations under different surface charges. Then the initial simulation box configuration was determined based on the density of ILs and the number of anions and cations inside the slit. The numbers of cations and anions in slits with different surface charge amounts and different coverage ratio are shown in Table 1. We also simulated the pure carbon slits with three layers (named as 3LC) of grahene sheets for comparison.

From Table 1, the number of cations and anions is equal in the slit of uncharged pure carbon. However, the number of anions exceeds that of cations due to attraction of TiO_2 to ions in the slits covered with carbon. And more anions are adsorbed on the surface of TiO_2 due to the small size of anions. In charged slits the number of cations and anions is unequal due to surface charge.

2.2. Preparation of the TiO_2/C composite materials

In order to verify the simulation results, we prepared TiO_2/C composite materials to be used to measure the EDL capacitance of TiO_2/C electrode in [emim][DCA]. The ionic liquid ([emim][DCA], purity 98%) was purchased from Lanzhou Institute of Chemical Physics of Chinese Academy of Sciences. TiO_2 (purity 99.8%, 25 nm, rutile) was purchased from Shanghai Aladdin Biotechnology Co., Ltd. YP-50F (Kuraray, Japan) is commercially available as a pure carbon material comparison for TiO_2/C composites. 4.0 g of TiO_2 and a certain amount of glucose (Sinopharm Chemical Reagent Co., Ltd.) and oxalic acid (Shanghai Lingfeng Chemical Reagent Co., Ltd.) were dissolved in absolute ethanol, wherein the mass ratio of glucose to oxalic acid is 96.5:3.5. For thorough mixing, the mixture was stirred with a magnetic stirrer for 12 h, and then the sample was heated at 90 °C in a drying oven for 5 h, and then heated at 150 °C for 5 h for the purpose of polymerization of glucose. The sample was then put in the tube furnace and heated to 800 °C under an argon atmosphere at a heating rate of 5 °C/min and maintained at this temperature for 3 h. Subsequently, it was taken out and ground to prepare the TiO_2/C composite and was used as working electrode material.

2.3. Material characterization and electrochemical testing

Transmission Electron Microscope (TEM, JEM-2100Plus) was used to observe the carbonization morphology. The final carbon content was determined from a thermogravimetric analyzer (TGA, NETZSCH STA 409 PC/PG). The specific surface area of the composite was tested by Brunner-Emmet-Teller (BET) measurements (BELSORP-MINI).

TiO_2/C composite material, acetylene black, polyvinylidene fluoride (PVDF, 1%) in 8:1:1 mass ratio were mixed into paste and coated on a carbon paper (TGP-H-060, Toray, Japan), which was then fixed to the working electrode (Shanghai Jingchong Electronic Technology Development Co., Ltd.). A sealed three-electrode glass electrolysis cell was used including the working electrode, a Pt (Shanghai Jingchong Electronic Technology Development Co., Ltd.) counter electrode and a Ag/AgCl (3 M KCl in the electrolyte) reference electrode (Shanghai Jingchong Electronic Technology Development Co., Ltd.). Many researchers have used three-electrode system with Ag/AgCl reference electrode to perform electrochemical measurement, and the measurement results show this measurement method is simple and effective[38–40]. [emim][DCA] was dissolved in acetonitrile in a glove box to obtain the mixed electrolyte with concentration of 0.25 M. In order to minimize the moisture in the electrolyte, ionic liquid and acetonitrile were pre-heated dehydration. Cyclic voltammetry (CV) were conducted by Autolab 302 N electrochemical workstation (Metrohm). The EDL capacitance of TiO_2/C electrode in [emim][DCA] can be calculated from CV curve.

3. Results and discussion

3.1. Number density distribution

In order to reveal the structure of the ILs in the graphene-coated TiO_2 slit, we analyzed the number density profiles of [emim][DCA], along the perpendicular to the graphene-covered TiO_2 plane (i.e. the z-axis), as shown in Fig. 3.

Fig. 3a shows the number density distribution of cation and anion in the uncharged slit. ILs show layered structure in the slit. The number density distribution of ions in slits of 100% coverage ratio of carbon and pure carbon is similar. But that in the slit with 49.5% and 78.7% coverage ratio of carbon is different. There are more peaks near the wall of the slits because the surface of TiO_2 which is not covered by carbon will absorb ions. In the charged slit the distribution changes. The layers of $[emim]^+$ and $[DCA]^-$ take place alternatively. There are more anions near the surface in the slit with surface charge of 0.06445 C/m^2, but opposite in the slit with surface charge of −0.06445 C/m^2. When the surface charge density goes up to 0.3222 C/m^2 or -0.3222 C/m^2, the layered structure is more obvious, and the regions of the anion and anion layers do not overlap. Atilhan et al. studied the interaction of [emim][DCA] ionic liquid with graphene, and found although the presence of graphene leads to layering, this does not weaken

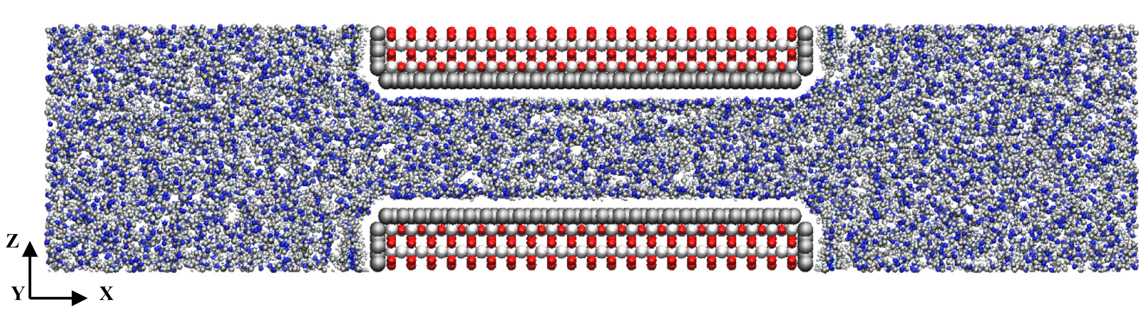

Fig. 2. Simulation box for initial density of ILs inside the slit.

Table 1
The number of cations and anions of ionic liquids in the slits with different surface charge densities(σ) and different coverage ratio of carbon.

σ(C/m^2)		0		0.06445		−0.06445		0.3222		−0.3222	
Ion name		[emim]$^+$	[DCA]$^-$	[emim]$^+$	[DCA]$^-$	[emim]$^+$	[DCA]$^-$	[emim]$^+$	[DCA]$^-$	[emim]$^+$	[DCA]$^-$
Coverage ratio of carbon	pure carbon (3LC)	213	213	205	227	218	197	186	294	262	159
	49.5%	232	268	226	282	237	252	218	342	276	214
	78.7%	216	252	211	269	222	236	198	337	262	188
	100%	206	243	200	258	213	226	184	330	255	175

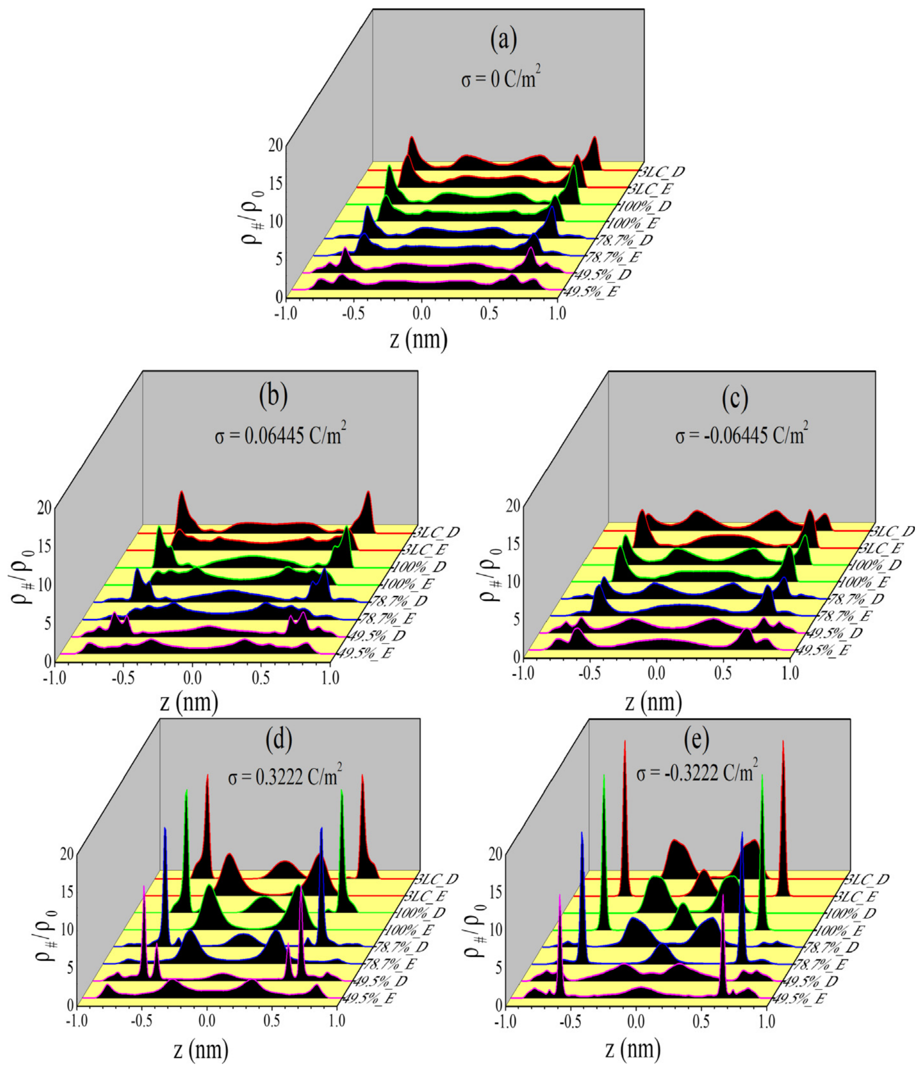

Fig. 3. Density distribution profiles of ionic liquids in the slits with different coverage ratio of carbon. (a) σ = 0 C/m^2 (b) σ = 0.06445 C/m^2 (c) σ = −0.06445 C/m^2 (d) σ = 0.3222 C/m^2 (e) σ = −0.3222 C/m^2 (The z-axis (parallel to the left wall) represents the coverage ratio of carbon. For example, 49.5%_E in the z-axis title represents density distribution of [emim]$^+$ in the slit with the coverage ratio of 49.5%, and 49.5%_D represents density distribution of [DCA]$^-$ in the slit with the coverage ratio of 49.5%).

remarkably anion−cation interactions because ion pairs are adsorbed onto the surface [35]. But when graphene is charged, the situation is different. From Fig. 3d and e the cation and anion layers are clearly separated, and this shows that the interaction between the anion and the anion can be weakened by the charged graphene. In the slits with 49.5% and 78.9% coverage ratio of carbon the surface of TiO_2 which is not covered by carbon will absorb both cations and anions. This effect will partly counteract the weakening of the interaction between the cation and the anion due to the charge and it should be beneficial to the stability of the electrode double layer. At this point, the surface heterogeneity may be beneficial to the capacitive performance of the electrode.

3.2. Orientation distribution

To further investigate the structure and properties of ILs in supercapacitors based on molecular level, we investigated the orientation properties of cations and anions, and the tilt angles of cations and anions are defined in Fig. 4. The orientation distribution probability of the cations and anions is shown in Fig. 5, and Fig. 6, respectively. It can be clearly seen that substantially most of the cations and anions have an tilt angle distribution of about 90°,

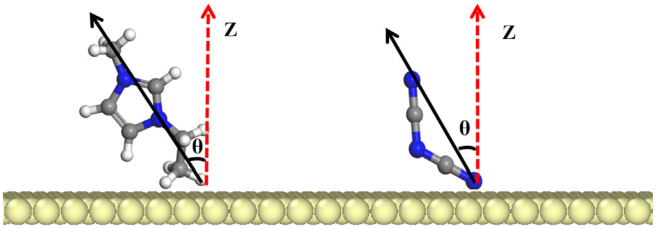

Fig. 4. Definition of the tilt angle of cation $[emim]^{+}$ and anion $[DCA]^{-}$. (the yellow slab represents the surface of slit).

which means that the ions tend to lie flat on the surface of the electrode. The orientation of ions on the electrode surface slightly changes with the coverage ratio of carbon and surface charge. The orientation distribution also reflects the number density distribution.

From Fig. 5a-d, when the charge applied to the surface is zero, the orientation of cations near the surface is parallel to the surface of slit. While the cations in the central region is relatively wide distributed. Its tilt angle is between 30° and 150°. The tilt angle distribution of cations near the surface is more dispersed in the slit with surface charge of 0.06445 C/m^2(Fig. 5e-h), but is more concentrated in the slit with surface charge of -0.06445 C/m^2(Fig. 5i-l). This is because the attraction of the negatively charged oxygen atom on the inner titanium oxide to the cation is counteracted by the surface positive charge in the slit with surface charge of 0.06445 C/m^2. While it is strengthened in the slit with surface charge of -0.06445 C/m^2. When the surface charge density goes up to 0.3222 C/m^2 or $^{-}$0.3222 C/m^2, the situation is about the same, but the layered structure is more obvious, the region of ion distribution is concentrated to the middle region, and the number of regions is reduced. The increase of coverage ratio (or the increase of carbon content) makes the cation distribution close to the middle region, and makes its orientation distribution wider.

In Fig. 6, the tilt angle distribution of anions is more concentrated at 90°. The surface negative charge makes the orientation distribution of anions more concentrated, while the surface positive charge weakens the concentration, namely, it makes the distribution more dispersed. Similarly, the increase of coverage ratio (or the increase of carbon content) makes the anions distributed close to the middle region, and makes its orientation distribution wider. Compared with other cases, the orientation distribution of anions in slit with 49.5% coverage ratio of carbon has some special features. And the distribution of anions near the surface of slit wall is more dispersed. This is because cations are repelled to the central region by the strong surface positive charge, and some anions near the surface of TiO_2 are aligned under the double interaction of cations and TiO_2.

3.3. Diffusion coefficient

IL is usually viscous, the ion transport is important during the charge/discharge process. Diffusion coefficient can be used to characterize the mobility of ions. Whether and how surface heterogeneity affects ion transport is an interesting topic. Fig. 7 shows the self-diffusion coefficient of $[emim]^{+}$ and $[DCA]^{-}$ with different graphene coverage ratio under different charging conditions (in terms of different surface charge densities.). For clarity, average diffusion coefficients in xy-directions and in z-direction are shown separately, the average diffusion coefficients of bulk ILs in xyz-direction are shown as dash line in the figure. In Fig. 7, all diffusion coefficients are smaller than those of bulk phase due to the effect of confinement. From Fig. 7a, the variation of coverage ratio of carbon does not have much impact to the diffusion coefficient of ions since the values of diffusion coefficient of the four cases are almost the same when the slits are uncharged. In the slits with surface charge density of 0.06445 C/m^2, the diffusion coefficient of the anion and cation is similar to the result of uncharged slit in xy-direction but the diffusion coefficient of cation is larger than that of anion in z-dirextion. When the surface charge density change to negative as -0.06445 C/m^2 (Fig. 3c), the diffusion coefficient of $[DCA]^{-}$ is higher than that of $[emim]^{+}$ in z-dirextion. This is reasonable because cations are adsorbed on the negatively charged

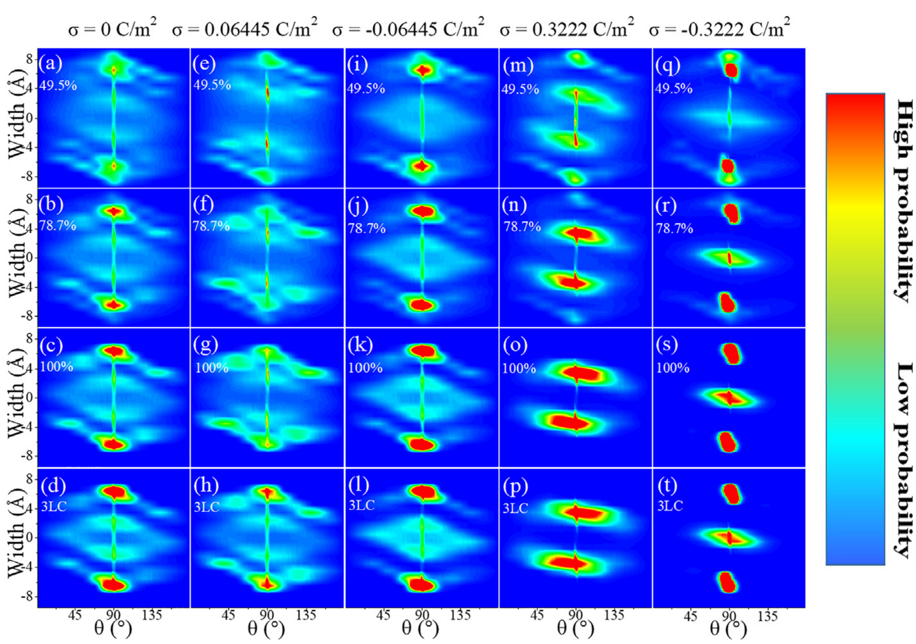

Fig. 5. Orientation distribution probability of cations in slits with different coverage ratio of carbon and different surface charge density. (a–d): $\sigma = 0$ C/m^2 with coverage ratio of 49.5%, 78.7%, 100%, 3LC, respectively. (e–h): $\sigma = 0.06445$ C/m^2 (i–l): $\sigma = -0.06445$ C/m^2 (m–p): $\sigma = 0.3222$ C/m^2 (q–t): $\sigma = -0.3222$ C/m^2 (Red represents high probability, and blue represents low probability).

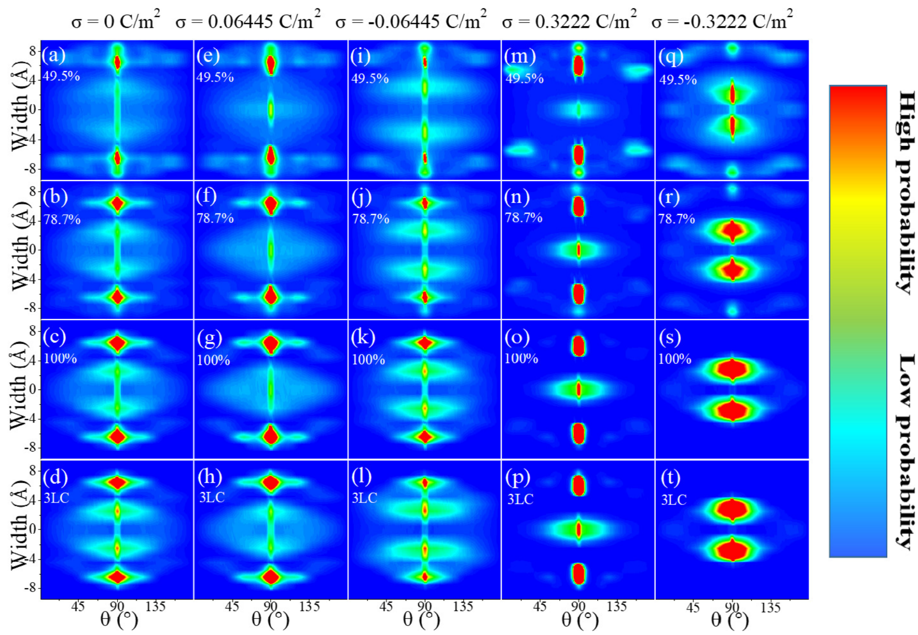

Fig. 6. Orientation distribution probability of anions in the slits with different coverage ratio of carbon and different surface charge density. (a–d): σ = 0 C/m^2 with coverage ratio of 49.5%, 78.7%, 100%, 3LC respectively. (e–h): σ = 0.06445 C/m^2 (i–l): σ = −0.06445 C/m^2 (m–p): σ = 0.3222 C/m^2 (q–t): σ = −0.3222 C/m^2 (Red represents high probability, and blue represents low probability).

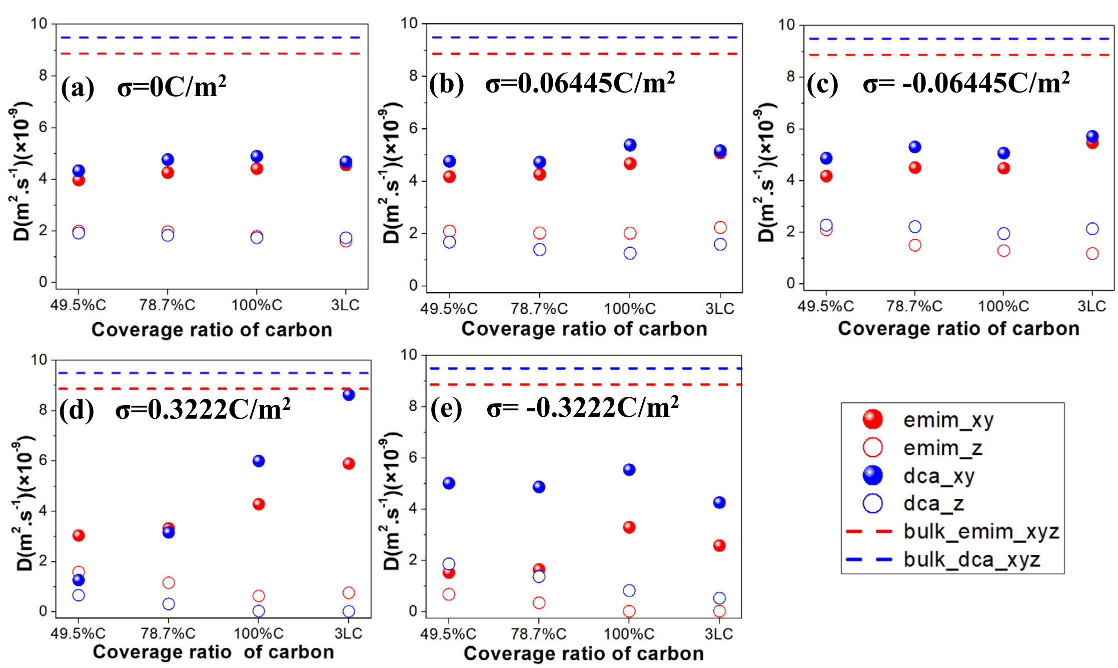

Fig. 7. The diffusion coefficient of [emim]$^+$ and [DCA]$^-$ in silts with different coverage ratio of carbon under different charging conditions (For clarity, average diffusion coefficients in xy-directions and in z-direction are shown separately, the average diffusion coefficients of bulk ILs in xyz-direction are shown as dash line in the figure) (a) σ = 0 C/m^2 (b) σ = 0.06445 C/m^2 (c) σ = −0.06445 C/m^2 (d) σ = 0.3222 C/m^2 (e) σ = −0.3222 C/m^2 (σ represents the surface charge density, emim_xy represents the average diffusion coefficient of [emim]$^+$ in the xy-direction, and emim_z represents the diffusion coefficient of [emim]$^+$ in the z-direction, and so on.)

surface and more confined on the surface, it reduces its mobility. When the surface negative charge density increases to −0.3222, cations are more strongly attracted, its diffusion coefficient becomes smaller not only in z-direction but also in xy-direction (Fig. 7e). But when the surface charge density goes up to 0.3222 C/m^2, it becomes very different. In Fig. 7d, the diffusion coefficient of [emim]$^+$ is larger than that of [DCA]$^-$ in xy-direction when the coverage of carbon is less than 100%, but for the silt with 100% coverage ratio or the pure carbon, the diffusion coefficient of [emim]$^+$ is lower than that of [DCA]$^-$. Normally, surface with positive charge will attract more anions, while the surface of TiO_2 which is not covered by carbon will adsorb ions. Meanwhile, a large number of adsorbed anions also attract cations. So the diffusion coefficient of both [emim]$^+$ and [DCA]$^-$ decrease in slits with partially covered. Because the anions are attracted more by the charged interface and TiO_2 surface, the diffusion coefficient of anion decreases more. But for silt with 100% coverage ratio of carbon and pure carbon, it is another case, where the interfacial layered structures of cations and anions is more obvious from the number density distribution profiles of Fig. 3d. The friction between layers decreases and the diffusion of ions mainly occurs inside the layer. The diffusion coefficient of anions and cations in Z direction is smaller, while that in xy-direction is larger. The diffusion coefficient of anions is larger because of their smaller geometry size and smaller steric hindrance. But why does the diffusion coefficient of $[emim]^+$ become smaller in slits with 100% coverage ratio of carbon and pure carbon under negative surface charge (Fig. 7e)? Because here the cation is near the wall of the slit, the large size of the cation limits its movement (Fig. 3e).

3.4. Integral capacitance

Chaban et al. [41] calculated the integral and differential capacitances as a function of the electrostatic potential, from the curves of capacitances they found the performance was best when the anions and cations were equal in size. Here we use integral capacitance to characterize the EDL of [emim][DCA] at heterogeneous interface. The electrostatic potential was computed from the charge distribution across the EDL inside the pore, and the charge distribution can be derived from number density distribution. Following Ma's paper[42], the electrostatic potential $\psi(z)$ can be derived by integrating Poisson Equation of equation (1).

$$\frac{d^2\psi}{dz^2} = -\frac{\rho e(z)}{\varepsilon} \tag{1}$$

$$\psi(z) = -\frac{1}{\varepsilon}\int_0^z (z-u)\rho e(u)du - \frac{\sigma}{\varepsilon}z \tag{2}$$

where, $\psi(z)$ is the electrostatic potential along z direction, $\rho_e(z)$ is the charge density, ε is the dielectric constant, $\varepsilon = \varepsilon_{r.}\varepsilon_0$, ε_0 is the vacuum dielectric constant, and ε_r is the relative dielectric constant of the electrolyte. Here $\varepsilon_r = 10$ according to Hunger's paper[35].

If we choose the potential at $z = 0$ (the surface of slit) to be zero (i.e., $\psi(0) = 0$), the total electrode potential $\Psi(surf)$ relative to the bulk area is calculated by equation (3). In the expression of the ψ_{Donnan} of equation (4), e is the elementary charge, $n_+(z)$, and $n_-(z)$ are the number density distributions of the cation $[emim]^+$ and the anion $[DCA]^-$, respectively. $\beta = 1/kT$, k is the Boltzmann constant, and T is the temperature. And σ is the surface charge density. Finally, the integral capacitance is obtained by equation (5).

$$\Psi(surf) = \psi_{Donnan} + \psi(0) \tag{3}$$

$$\psi_{Donnan} = -\frac{1}{\beta e}\ln\sqrt{\frac{\frac{1}{L}\int_0^L n_+(z)\exp(\beta e\psi(z))dz}{\frac{1}{L}\int_0^L n_-(z)\exp(-\beta e\psi(z))dz}} \tag{4}$$

$$C = \frac{\sigma_+ - \sigma_-}{\Psi_+(surf) - \Psi_-(surf)} \tag{5}$$

Fig. 8 shows the charge density distribution in slits with different graphene coverage ratio. The charge density actually represents the difference between the local densities of cations and anions. When the surface is positively charged, most of anions are adsorbed near the surface and negative charge distribution peak appears and vice versa. The peak height increases with the increase of surface charge density. Even for the surface with zero charge (Fig. 8a), the slit partially covered with carbon has a small negative charge distribution peak close to the surface (about 1.15 Å away from the surface). When the surface charge density is 0.06445 C/m^2 (Fig. 8b) or -0.06445 C/m^2 (Fig. 8c), this small peak still exists. This is caused by TiO_2, which is not covered with carbon. TiO_2 adsorbs some ions, which leads to the decrease of the number of ions near the middle region, and results in a decrease in amplitude of peaks near the surface (about 3.15 Å away from the surface) and in the middle region. The charge density distribution in the middle region of pure carbon slit with surface charge density of 0.06445 C/m^2 is similar a straight line. This is mainly due to the fact that the anions and cations are mixed together without obvious boundary. And this can also be seen from Fig. 4. Since middle layer of charge acts to reduce the EDL screening of the pore charges and lowers the capacitance, the capacitance of pure carbon slit with surface charge density of 0.06445 C/m^2 will has higher capacitance compared with other slits because it has no charge distribution in the middle region and lower positive charge peak near the negative charge peak. This can be confirmed by the potential distribution of Fig. 9d and the integral capacitance of Fig. 10. In Fig. 8c, in addition to the titanium oxide adsorption peak, the first high peak near the surface is a negative charge peak, which cannot play a screening role, but will reduce the screening role of the second positive charge peak. The pure carbon slit has lower negative peak and higher positive peak compared with other slits with surface charge density of 0.06445 C/m^2, its capacitance will be higher. From the above discussion TiO_2 reduce the EDL screening of the pore charges and lowers the capacitance at low Potential. However, when the surface positive charge density increases to 0.3222 C/m^2 or negative charge density increases to -0.3222 C/m^2, the adsorption peak of TiO_2 is not so obvious. In Fig. 9d, the high screening peaks of the slit covered with carbon are higher than that of the pure carbon slit. Although the high screening peak of the pure carbon slit is higher than that of other slits in negative slits (Fig. 8e), its peaks of middle layer of charge are also higher, which partly reduce the screening of EDL. Therefore the capacitance of pure carbon slit will decrease at high electrode potential (e.g. high surface charge density).

The potential distribution can be calculated from the charge density with equation (7). Fig. 9 shows the electrostatic potential distribution in the slits with different coverage ratio of carbon and pure carbon. Even for the surface with zero charge (Fig. 9a), the slit partially covered with carbon has a positive potential distribution, this is because even on uncharged surfaces, ILs can form DEL. Due to the existence of TiO_2, the number of anions near the surface increases, and the charge distribution is obviously negative. Given the surface potential is $\psi(0) = 0$, the potential distribution $\psi(z)$ is positive according to equation (7). TiO_2 adsorption layer even leads to 'over screening' in Fig. 10b. When the surface has a positive charge density of 0.06445 C/m^2, the potential distribution of slits covered with carbon still positive, while the pure carbon slit has negative potential distribution (Fig. 9a). This phenomenon is conducive to the increase of capacitance. However, when the surface has a negative charge density of -0.06445 C/m^2, it does the opposite and reduces the screening (Fig. 9c). The final capacitance depends on the balance of the two slits. When the surface charge density increases to 0.3222 C/m^2, the adsorption of ions on TiO_2 is weakened, but the potential distribution is 'softened' by the TiO_2, the potential distribution of silt covered carbon is flatter than that of pure carbon slit (Fig. 9d). It's helpful for screening effect. However, it has the opposite effect for the negative slit (Fig. 9e).

Fig. 10 shows the integral capacitance according to Eq. (5). As discussed above the capacitance of pure carbon slit with surface charge density of 0.06445 C/m^2 has higher capacitance compared with other slits because it has no charge distribution in the middle region and lower positive charge peak near the negative charge

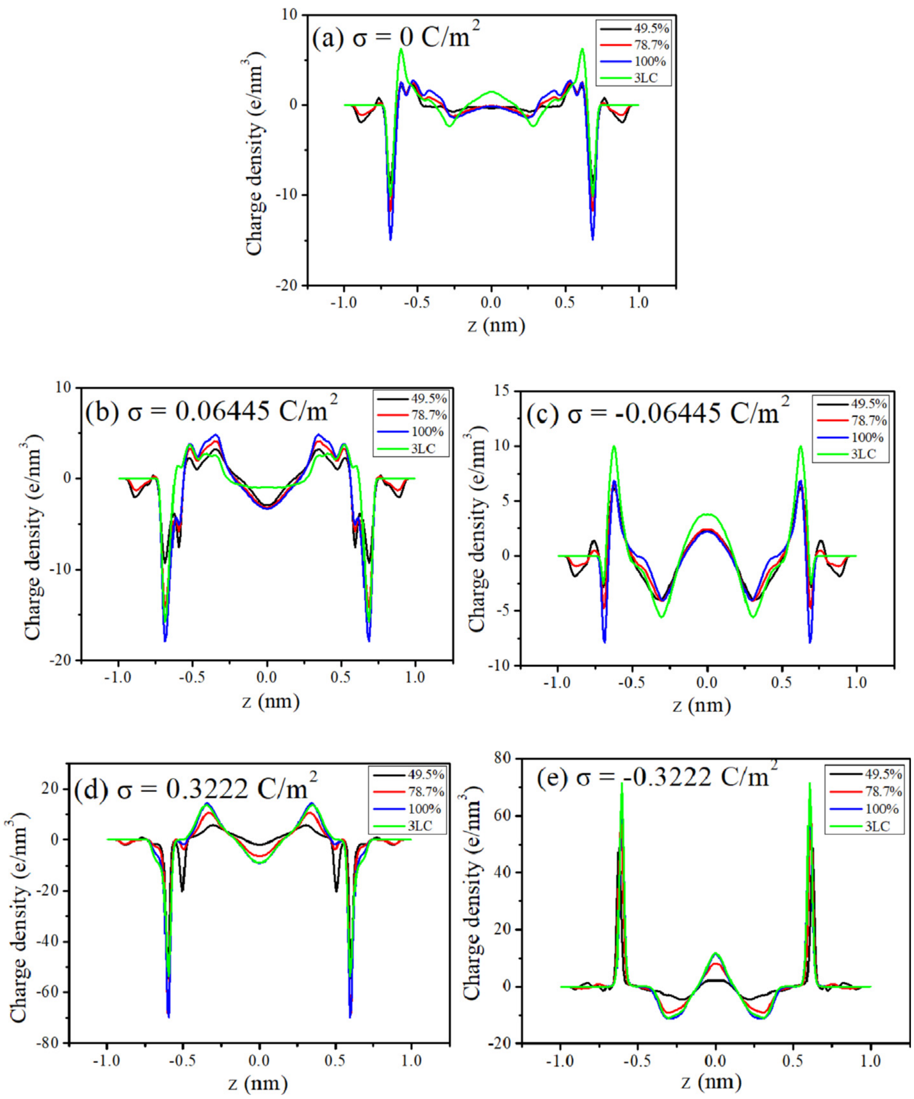

Fig. 8. Charge density distribution in slits with different coverage ratio of carbon. The surface charge density is (a): $\sigma = 0$ C/m^2, (b): $\sigma = 0.06445$ C/m^2, (c): $\sigma = -0.06445$ C/m^2, (d): $\sigma = 0.3222$ C/m^2, (e): $\sigma = -0.3222$ C/m^2, respectively.

peak (Fig. 8a). However, when the charge density increases to 0.3222 C/m^2, the capacitance of pure carbon slit decreases dramatically. It can be seen from the density distribution (Fig. 4d, e) that the boundary between cation layer and anion layer becomes obvious, and the co-ions in the middle region reduce the screening effect of counter ions. In this way, the anti-screening effect of co-ions reduces the capacitance of pure carbon slit at high potential. On the other hand, the existence of TiO_2 softens the stratification of counter ions and co-ions, and the slits partially carbon covered have higher capacitance. Considering the necessity of charging and discharging at high electrode potential the slit of TiO_2 partially covered with carbon is more suitable for capacitor. From Fig. 10 the slit with coverage ratio of 78.7% has the best performance of capacitance at high electrode potential.

3.5. Experimental results

The simulation results show that the material with inhomogeneous surface by partially covered carbon may have good capacitance performance. In order to verify the simulation results, we prepared TiO_2/C composite materials with different coverage ratio of carbon using different amounts of glucose to carbonize TiO_2. The microstructure of the material is characterized by TEM. As shown in Fig. 11, there are some amorphous materials around the crystal particles. And we can infer that TiO_2 nanoparticles of all samples were coated with carbon since the TiO_2 is rutile crystal with purity of 99.8%. The TGA of TiO_2/C composites with different carbon contents is shown in Fig. 12, the loss of major mass begins at 400 °C and stabilizes at 600 °C. The carbon content of each composite material was 4.79%, 6.70%, 11.89%, 13.1% by calculating weight loss rate and expressed as TiO_2/0.0479C, TiO_2/0.0670C, TiO_2/0.1189C, TiO_2/0.1310C, respectively.

The measurement of the specific surface area (S_{BET}) of the TiO_2/C composite material is performed by BET method. And the specific surface area of each material is shown in Table 2. YP-50F is a pure carbon material as a comparison for TiO_2/C composites. All of the TiO_2/C composite material has low specific surface area compared with that of commercial carbon of YP-50F. The low specific surface area of TiO_2/C composite is understandable because no activation is carried on the material to increase its surface area. Since our goal is to figure out the effects of surface heterogeneity, the influence of different area on the capacitance value can be eliminated through

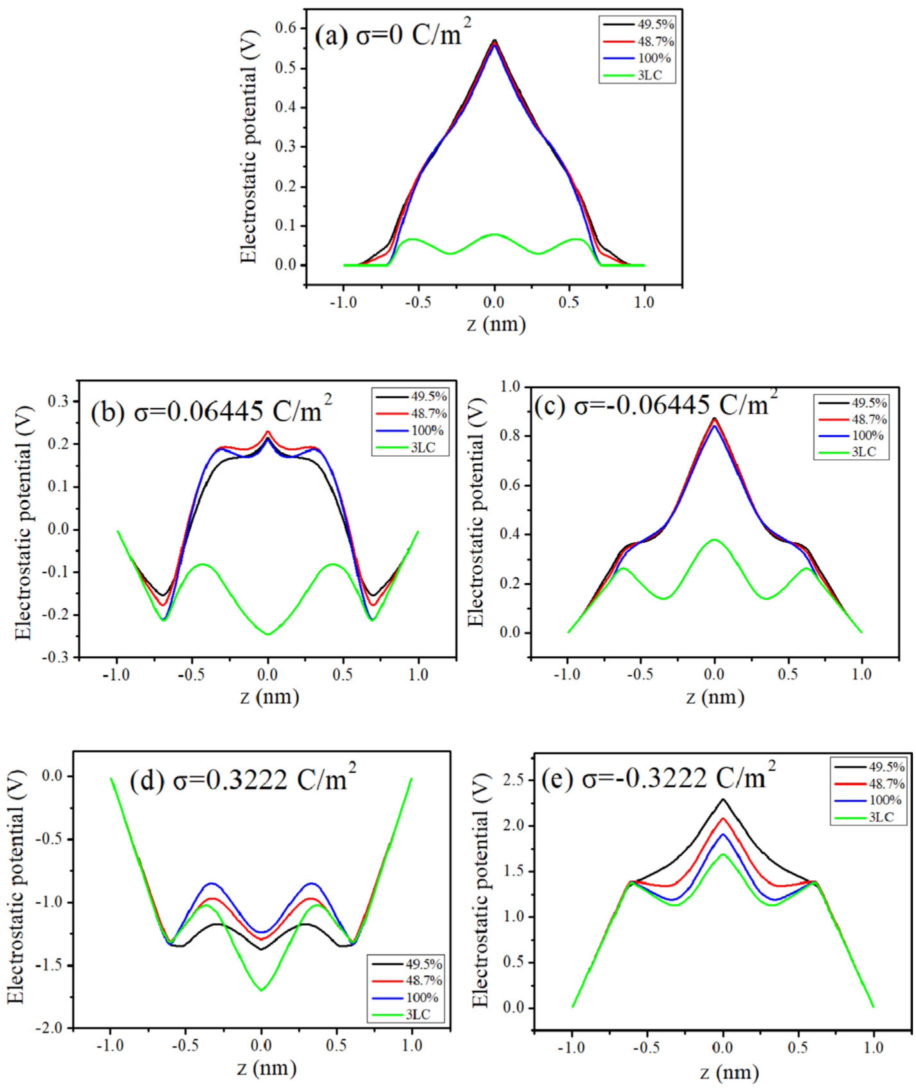

Fig. 9. Electrostatic potential distribution in slits with different coverage ratio of carbon. The surface charge density is (a): σ = 0 C/m^2, (b): σ = 0.06445 C/m^2, (c): σ = −0.06445 C/m^2, (d): σ = 0.3222 C/m^2, (e): σ = −0.3222 C/m^2, respectively.

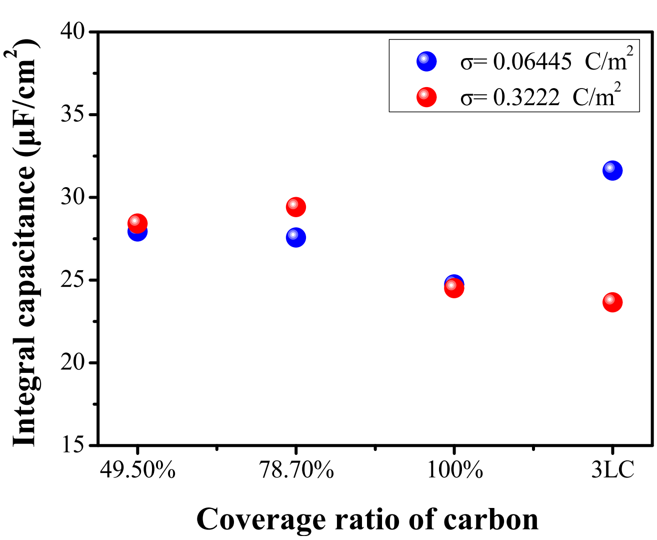

Fig. 10. Integral capacitance of ILs in slits with different coverage ratio of carbon.

dividing the capacitance value by their specific surface area.

The CV curves of materials with different carbon contents are shown in Fig. 13 with scanning rate of 100mv/s. The curves of TiO_2/C composites are all quasi-rectangular, which reflects the good electric double layer properties of the composite. Although the specific current of YP-50F is higher than that of the composite of TiO_2/C in positive scan range from −0.44 V to 0 V and reverse scan range from −0.14 V to −0.6 V, there is a significant current decrease in the scan regions. This may be due to the slow surface diffusion of the electrolyte to the material since pure carbon materials are hydrophobic, while the composite of TiO_2/C has partial hydrophilicity as the heterogeneous material.

Based on the CV curve the specific areal capacitance can be calculated according to the equations (6) and (7):

$$C_p = \frac{1}{mv(V_2 - V_1)} \int_{V_1}^{V_2} I dV \tag{6}$$

$$C_S = \frac{C_p}{S_{BET}} \tag{7}$$

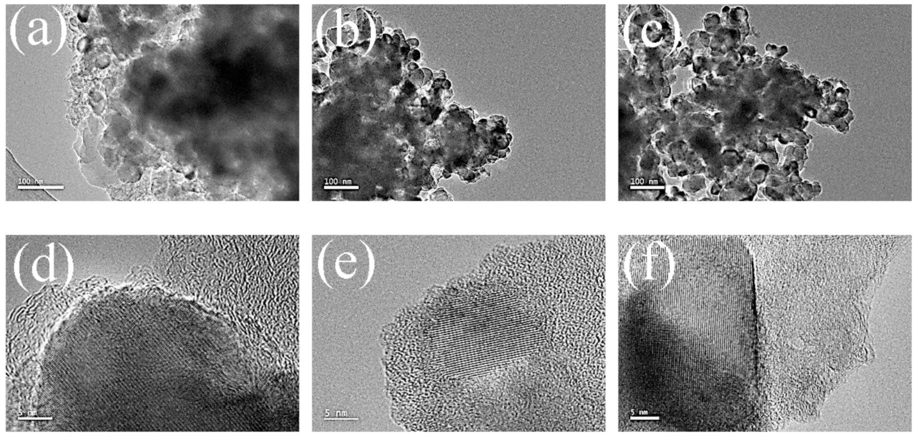

Fig. 11. TEM images of TiO_2/C composites and TiO_2. (a–c) are images of TiO_2/0.0670C, TiO_2/0.1189C, and TiO_2/0.1310C with resolution of 100 nm, respectively. (d–f) are images of TiO_2/0.0670C, TiO_2/0.1189C, and TiO_2/0.1310C with resolution of 5 nm.

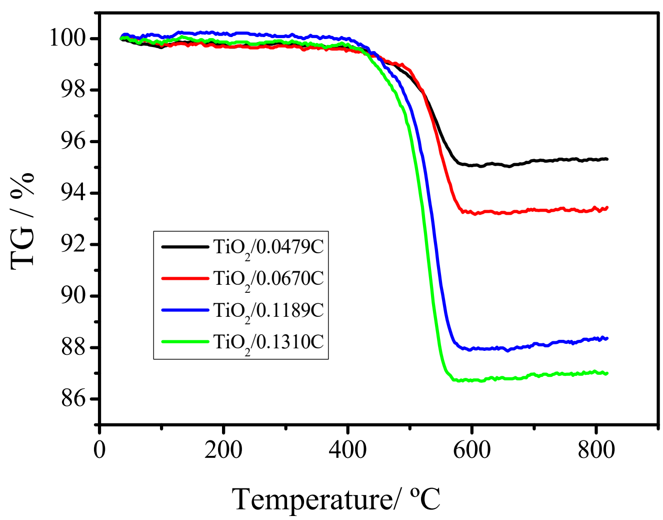

Fig. 12. TGA of TiO_2/C composites with different carbon contents.

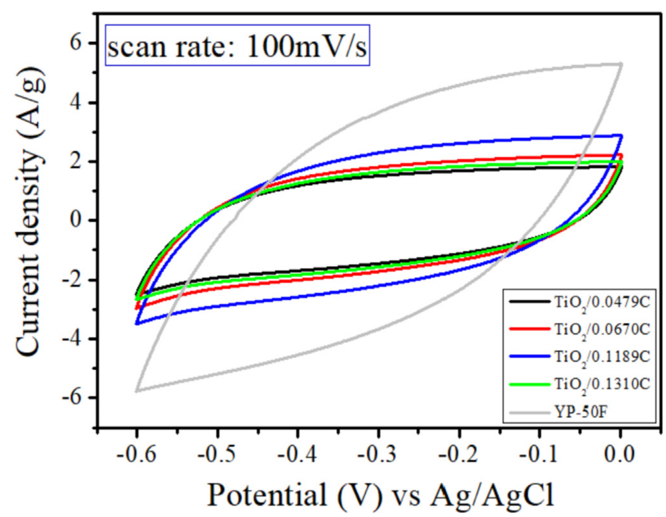

Fig. 13. CV curve of materials with different carbon contents in 0.25 M [emim][DCA] electrolyte, scan rate is 100 mV/s.

Table 2
BET characteristics of composites with different carbon contents.

	C % (m/m)	S_{BET} (m^2/g)
TiO_2/0.0479C	4.79	40.28
TiO_2/0.0670C	6.70	44.38
TiO_2/0.1189C	11.89	73.72
TiO_2/0.1310C	13.10	94.34
YP-50F	100	1600

Here, C_p is the specific mass capacitance, m is the mass of the active material in the working electrode, v is the scanning rate, I is the current during the test, and V_1 and V_2 are the low voltage value and the high voltage value of the voltage window, respectively. C_s is the specific area capacitance and S_{BET} is the specific surface area of the TiO_2/C composites. Because the specific area capacitance considers the influence of specific surface area, and its unit is μF/cm^2, which is consistent with the unit of specific capacitance obtained by simulation, experimental results can be directly compared with simulation results through specific areal capacitance.

The specific capacitance of material with different carbon contents is shown in Fig. 14. The material of TiO_2/0.1189C has the highest performance of specific areal capacitance from Fig. 14b. Although YP-50F has the highest specific mass capacitance, considering its much larger specific surface area than the composite, it does not have an advantage in capacitance performance. If the specific mass capacitance is divided by the specific surface area, and the capacitance value is presented by the specific areal capacitance, the capacitance performance of YP-50F is much lower than that of the composite. The composite of TiO_2/0.0670C is the material with the optimal carbon content; its specific areal capacitance is high up to 31.53 μF/cm^2 and 19 times the value of YP-50F of 1.64 μF/cm^2.

The existence of the optimal carbon content is qualitatively consistent with the optimal carbon coverage ratio obtained by simulation. These results suggest that heterogeneous materials may be a good choice for capacitor materials. It should be noted that the materials we prepared have not been activated, and its specific surface area is very low. Therefore, the material preparation method in this paper cannot be used as capacitor material directly. Nevertheless, we have proved that the heterogeneous surface can promote the capacitance of the EDLC. This is of great significance for the experimental synthesis of high performance capacitor materials.

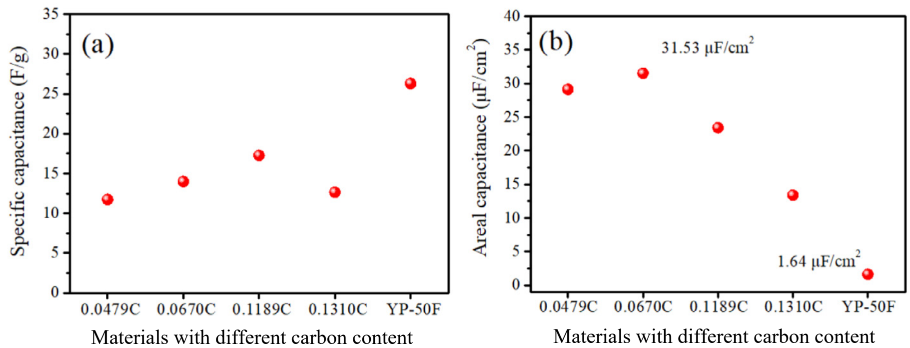

Fig. 14. Specific mass capacitance(a) and specific areal capacitance(b) of material with different carbon contents. (0.0479C, 0.0670C, 0.1189C, 0.1310C represent TiO_2/0.0479C, TiO_2/0.0670C, TiO_2/0.1189C, TiO_2/0.1310C, respectively.)

4. Conclusions

We investigated the electric double-layer of [emim][DCA] on slit pore of TiO_2/C with the heterogeneous surface and different coverage ratio of carbon by molecular dynamics simulation. Meanwhile, we prepared TiO_2/C composite materials with different content of carbon using different amounts of glucose to carbonize TiO_2 and performed experimental measurements.

The results of simulation show that coverage ratio of carbon on the surface affect the density distribution, orientation and diffusion coefficient of [emim][DCA]. In the slits with partly coverage ratio of carbon the surface of TiO_2 which is not covered by carbon will adsorb ions. This effect should be beneficial to the stability of the electrode double layer. The increase of coverage ratio (or the increase of carbon content) makes the cation distribution close to the middle region, and makes its orientation distribution wider. The effect of coverage ratio of carbon on the orientation distribution of anions is similar to that of cations. The variation of coverage ratio of carbon does not have much effect on the diffusion coefficient of ions when the slits are uncharged. But the case is different in charged slit. Especially, when the surface charge density goes up to 0.3222 C/m^2, it becomes very different. The diffusion coefficient of $[emim]^+$ is larger than that of $[DCA]^-$ when the coverage ratio of carbon is less than 100%, but for the silt pore with 100% coverage ratio or the pure carbon, the diffusion coefficient of $[emim]^+$ is lower than that of $[DCA]^-$. It shows that the coverage ratio has a great influence on the diffusion coefficient of anion and anion when the slit surface has a large charge. Moreover, we calculated the charge distribution of electric double-layer of [emim][DCA] and the capacitance. The results indicate the capacitance of slit pore with the surface of Titania with 78.7% carbon coverage ratio has the highest value at high electrode potential.

We performed an electrochemical experiment using composite of TiO_2/C as the electrode material, and obtained the capacitance from cyclic voltammetry (CV). The experimental results also indicate that there is an optimal coverage ratio of carbon, which achieves the highest capacitance. This shows that the experimental results qualitatively verified the prediction of the simulation. We have proved that the heterogeneous surface can promote the capacitance of the EDLC by molecular simulation and experiment. This is of great significance for the experimental synthesis of high performance capacitor materials. Our researches could provide a microscopic perspective of the electric double layer at the heterogeneous interface and the guidance for the design of electrode material.

Declaration of competing interest

There are no conflicts of interest to declare.

CRediT authorship contribution statement

Jiabao Zhu: Investigation, Methodology, Writing - original draft. **Linghong Lu:** Resources, Writing - review & editing, Supervision, Data curation. **Lili Shi:** Validation, Formal analysis. **Zhongyang Dai:** Visualization, Software. **Wei Zhuang:** Writing - review & editing. **Zhengsong Weng:** Methodology.

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